

Lingyu Zhang

Ph.D. Student in ECE, Duke University
[Google Scholar](#) · [GitHub](#) · [Email](#) · [Website](#)

RESEARCH INTERESTS

Embodied AI, Foundation Models & Policy Adaptation

I study how foundation models and human feedback can accelerate learning in embodied agents.

EDUCATION

Duke University - Expected 2028

Ph.D. Electrical & Computer Engineering
Advisor: Boyuan Chen

Columbia University - 2023

M.S. Electrical Engineering

Nanjing University - 2021

B.E. Microelectronic Science & Engineering

SELECTED PUBLICATIONS

Full publication list [Google Scholar](#)

7. RTNav: Towards Real-time Zero-Shot Object Navigation. Preprint 2026.
Lingyu Zhang*, Easop Lee*, Boyuan Chen
[paper](#) · [code](#) · [project](#) · [video](#)
6. Enabling Multi-Robot Collaboration from Single-Human Guidance, **ICRA** 2025.
Zhengran Ji, **Lingyu Zhang**, Paul Sajda, Boyuan Chen
[paper](#) · [code](#) · [project](#) · [video](#)
5. Scientific Judgment Drifts Over Time in AI Ideation, Preprint 2025.
Lingyu Zhang, Mitchell Wang, Boyuan Chen
[paper](#) · [code](#) · [project](#) · [video](#)
4. GUIDE: Real-Time Human-Shaped Agents, **NeurIPS** 2024.
Lingyu Zhang, Zhengran Ji, Nicholas R Waytowich, Boyuan Chen
[paper](#) · [code](#) · [project](#) · [video](#)
3. CREW: Facilitating Human-AI Teaming Research, **TMLR** 2024.
Lingyu Zhang, Zhengran Ji, Boyuan Chen
[paper](#) · [code](#) · [project](#) · [video](#)
2. Adversarially Robust Video Perception by Seeing Motion, Preprint 2023.
Lingyu Zhang*, Chengzhi Mao*, Junfeng Yang, Carl Vondrick
[paper](#) · [project](#)
1. Robust Perception through Equivariance, **ICML** 2023.
Chengzhi Mao, **Lingyu Zhang**, Abhishek Vaibhav Joshi, Junfeng Yang, Hao Wang, Carl Vondrick
[paper](#) · [code](#) · [project](#)

ONGOING RESEARCH

Foundation Models for Policy Transfer Across Extreme Visual Fidelity Gaps

Investigating whether foundation model representations can enable policy transfer from low-fidelity environments to visually realistic environments.

TECHNICAL SKILLS

Machine Learning: PyTorch, JAX, Hugging Face Transformers, vLLM

Robotics & Simulation: MuJoCo, Habitat-lab/-sim, Unity, LIBERO/robosuite

Tools: Docker, Weights & Biases, Hydra, Git, Linux

RESEARCH EXPERIENCE

Graduate Researcher, Duke University 08/2023 - Present

Foundation model & human-guided embodied learning.

Advisor: Boyuan Chen

Research on foundation-model-driven embodied policies, real-time human feedback for RL, and LLM-driven scientific ideation.

Graduate Research Assistant, Columbia University

Vision-Language Understanding & Fact Verification 02/2023 - 08/2023

Advisor: Shih-Fu Chang, Mentor: Mingyang Zhou

Studied VLMs for text-rich image understanding and multimodal fact verification.

Robust Computer Vision Models.

02/2022 - 11/2022

Advisors: Carl Vondrick & Junfeng Yang, Mentor: Chengzhi Mao

Incorporating equivariance and motion consistency to improve vision model robustness.

Undergraduate Researcher, Nanjing University

Learned Image Compression.

12/2020 - 06/2021

Advisor: Qiu Shen

Studied multiscale encoder architectures for end-to-end learned neural image compressors.

ACADEMIC ACTIVITIES

Peer Review

CoRL 2025-2026, Neurips 2026

Mentorship

Zhengran Ji - Human-AI/Robot Teaming

Mitchell Wang - AI-driven Scientific Ideation

Teaching Assistant

Robot Learning - Fall 2025, Duke University

Being a Successful ECE Master's Student - Fall 2026, Duke University

LEADERSHIP

PhD Representative ECE, Engineering Graduate Student Council, Duke 2025-2026

Career Chair, Graduate Electrical Engineering Council, Columbia 2022

Lead Department Rep, Engineering Graduate Student Council, Columbia 2022

President, Nanjing University Hip-Hop Association 2018-2019

SELECTED PRESS COVERAGE

Interesting Engineering, "Tiny Robots Team Up Like Humans with 84% Mission Success, Learn to Ambush, Encircle," 2025.

Tech Explorist, "Researchers Develop a New Framework to Train AI in Real-Time," 2024.

Open Source For You, "Open-Source Platform for Human-AI Teaming Playground," 2024.

Duke Engineering, "Training AI Through Human Interactions Instead of Datasets," 2024.

Duke Engineering, "A Launch Pad for Human-AI Teaming," 2024.